```
x=5
y=9
print("Two values are {1} and {0}".format(x,y))
print(f'Two values are {x} and {y}')
```

```
Two values are 9 and 5
Two values are 5 and 9
```

```
import random
random.randint(1,10)
```

```
3
```

```
print('I teach {1} and {0}'.format('Python','IoT'))
print('The examples of dictionary are {} and {}'.format(dict1, dict2))
```

```
I teach IoT and Python
The examples of dictionary are {1: 'value', 'key': 2} and {'Sagar': 'Asst. Prof.', 'SIT': 'College', 'Amit': 'Asst. Pr
```

```
num = input("Enter a Number: ")
print(num)
```

```
Enter a Number: 5
5
```

Operators in Python:

```
Arithmetic: +, -, /, *, %, // (Floor Division), ** (exponent operator)
Comparison: <, >, ==, !=, >=, <=
Logical: and, or, not
Bitwise: |, &, ~, ^, >>, <<
Assignment: =, +=, -=, *=, /=, //=, %=, **=, |=, &=, ^=, <<=, >>=
Identity: is, is not
Membership: in, not in
Ternary Operator: [on true] if(exp) else [on false]
```

```
x = 5
#x += 5
print(x*5)
print(8<<2)
letter = input('Enter a Character')
print('Vowel') if letter in 'aeiou' else print('Not Vowel')
```

```
25
32
Enter a Charactere
Vowel
```

```
print(~-2)
m = 5.5
n = int(m)
print(m==n)
```

```
#Ternary Operator in Python: [on true] if(exp) else [on false]
a = int(input("Enter 1st number: "))
b = int(input("Enter 2nd number: "))
min = a if(a<b) else b
print(min)
```

```
print("Sagar" in dict2)
print("BIT" not in dict2)
```

```
H = {'A': 'Apple', 'B': 'Biscuit', 'C': 'Cup', 'D': 'Duck', 'E': 'Envelope'}
'C' in H
```

```
Friends = ['sagar', 'amit', 'amol', 'kamlesh', 'kapil']
print('amit' in Friends)
print('kapil' not in Friends)

x = input()
y = input()
z = input()
print (x, 'is the largest number') if x>y and x>z else print (y, 'is the largest number') if y>z else print (z, 'is the la
```

```
x = input()
y = input()
z = input()
print (x, 'is the largest number') if x>y and x>z else print (y, 'is the largest number') if y>z else print (z, 'is the la
```

```
#Generate Random Numbers in a range
import random
print(random.randint(1,69))
```

Double-click (or enter) to edit

Double-click (or enter) to edit

Control Statements

selection statements

if statement
if ... else statement
if ... elif ... else statement

iteration statements

while statement (repeats as long as the condition is true)
Note: Use while when number of iteration are not known/fixed
for statement (repeats for every action in a sequence of items)
Note: Use for when number of iterations are known/fixed

```
##if
if 1 > 5:
  print('7 is Greater')
  print("Hello World!")
print(7*7)
print(3+5)
```

```
49
8
```

```
x = int(input("Enter Num 1: "))
y = int(input("Enter Num 2: "))
z = int(input("Enter Num 3: "))
L = x
if L<y:
  L = y
if L<z:
  L = z
print(f"Largest is {L}.")
print("Largest is ", L)
print(f"The Largest among {x}, {y}, {z} is {L}.")
```

```
Enter Num 1: 5
Enter Num 2: 8
Enter Num 3: 4
Largest is 8.
Largest is  8
The Largest among 5, 8, 4 is 8.
```

```
##if ... else
lst2 = [32, 45, 64, 37, 85]
x = int(input("Enter Element to be searched: "))
if x in lst2:
    print("Element found")
else:
    print("Element not found")

print("SIT")
```

```
Enter Element to be searched: 37
Element found
SIT
```

```
num1 = int(input("Enter a number: "))
if num1 > 30:
  print(f"{num1} is greater than 30")
else:
  if num1 == 30:
    print(f"{num1} is equal than 30")
  else:
    print(f"{num1} is less than 30")
```

```
Enter a number: 21
21 is less than 30
```

```
##if ... elif ... else
x = int(input("Enter any Number: "))
if ( x > 30 ):
  print("Hi, greater than 30")
elif ( x < 30 ):

  print("Hi, smaller than 30")
else:
    print("x = 30")
```

```
Enter any Number: 30
x = 30
```

```
x = int(input("Enter any Number: "))
if x > 100:
  print("Invalid Marks")
elif x > 90:
  print("A+")
elif x > 80:
  print("A")
elif x > 70:
  print("B+")
elif x > 60:
  print("B")
elif x > 50:
  print("C+")
elif x > 40:
  print("C")
else:
  print("Fail")
```

```
Enter any Number: 53
C+
```

```
##for using range
for i in range(10):
  print(i)
```

```
0
1
2
3
4
5
6
7
8
9
```

```
##for using range
#The range() function returns a sequence of numbers, starting from 0 by default,
#and increments by 1 (by default), and stops before a specified number.
#range(start, stop, step)
for i in range(20,0,-2):
  print(i)
str1='aabbjkhg'
str1[-3::1]
```

```
20
18
16
14
12
10
8
6
4
2
'khg'
```

Double-click (or enter) to edit

```
x = range(3, 6)
for n in x:
  print(n)
```

```
3
4
5
```

```
for i in range(10):
  print(i+1, "Symbiosis Institute of Technology, Nagpur")
```

```
1 Symbiosis Institute of Technology, Nagpur
2 Symbiosis Institute of Technology, Nagpur
3 Symbiosis Institute of Technology, Nagpur
4 Symbiosis Institute of Technology, Nagpur
5 Symbiosis Institute of Technology, Nagpur
6 Symbiosis Institute of Technology, Nagpur
7 Symbiosis Institute of Technology, Nagpur
8 Symbiosis Institute of Technology, Nagpur
9 Symbiosis Institute of Technology, Nagpur
10 Symbiosis Institute of Technology, Nagpur
```

```
Friends = ['Sagar', 'Amol', 'Roshan', 'Akshay', 'Sachin']
for frnd in Friends:
  print(frnd,":", len(frnd))
```

```
Sagar : 5
Amol : 4
Roshan : 6
Akshay : 6
Sachin : 6
```

```
##for using list
Friends = ['sagar', 'amit', 'amol', 'kamlesh', 'kapil']
for friend in Friends:
  print(f"The length of {friend} is {len(friend)}" )
```

```
The length of sagar is 5
The length of amit is 4
The length of amol is 4
The length of kamlesh is 7
The length of kapil is 5
```

```
##for using dictionary
bears = {"Grizzly":"angry", "Brown":"friendly", "Polar":"friendly"}
for bear in bears:
    if bears[bear]=="friendly":
        print("Hello, "+bear+" bear!")
    else:
      print("Bad Bear")
```

```
Bad Bear
Hello, Brown bear!
Hello, Polar bear!
```

```
num = int(input("enter your range:"))
a = 1
for i in range(num+1):
  if i == 2:
    print(i,"prime number")
  elif i == 1:
    print(i,"prime number")
  elif i == 3:
    print(i,"prime number")
  elif i == 5:
    print(i,"prime number")
  elif i == 7:
    print(i,"prime number")
  else:
    if i%2 == 0:
      a +=1
    elif i %3==0:
      a +=1
    elif i%5 == 0:
      a +=1
    elif i%7 ==0:
      a +=1
    else:
        print(i,"prime number")
print("none prime number =",a)
```

```
enter your range:100
1 prime number
2 prime number
3 prime number
5 prime number
7 prime number
11 prime number
13 prime number
17 prime number
19 prime number
23 prime number
29 prime number
31 prime number
37 prime number
41 prime number
43 prime number
47 prime number
53 prime number
59 prime number
61 prime number
67 prime number
71 prime number
73 prime number
79 prime number
83 prime number
89 prime number
97 prime number
none prime number = 76
```

```
start = int(input("Enter the start of the range: "))
end = int(input("Enter the end of the range: "))

print("Prime numbers between", start, "and", end, "are:")

for num in range(start, end + 1):
    if num > 1:
        is_prime = True
        for i in range(2, num):
            if (num % i) == 0:
                is_prime = False
                break
        if is_prime:
            print(num)
```

```
Enter the start of the range: 1
Enter the end of the range: 20
Prime numbers between 1 and 20 are:
2
3
5
7
11
13
17
19
```

```
#Loops with an else statement
#it is executed when the loop terminates through exhaustion of the
#iterable (with for) or
#when the condition becomes false (with while), but not when the
#loop is terminated by a break statement.
num=int(input('Enter positive integer: '))
for i in range(2,num):
    if num%i == 0:
        print("{} is not prime".format(num))
        break
else:
    print("{} is prime".format(num))
```

```
Enter positive integer: 3
3 is prime
```

```
rows = 5
for i in range(0, rows):
    for j in range(0, i + 1):
        print("*", end=' ')
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

```
for i in range(rows+1, 0, -1):
    for j in range(0, i - 1):
        print("*", end=' ')
    print()
```

```
* * * * *
* * * *
* * *
* *
*
```

```
##while
num = int(input("Enter a Number: "))
num1, sum = num, 0
while(num1!=0):
  sum += num1%10
  num1 //= 10 #This is the floor division operator. It performs integer division, discarding any remainder.
  #let's say num1 initially holds the value 123. After executing num1 //= 10,
  #the value of num1 will become 12, as 123 // 10 equals 12 in integer division. This process continues in the loop until
print(f"The sum of digits of {num} is {sum}")
```

```
Enter a Number: 123
The sum of digits of 123 is 6
```

```
# Armstrong number is a number that is equal to the sum of its own digits each raised to the power of the number of digits
#The number of digits in 153 is 3.
#Each digit of 153 raised to the power of 3 (the number of digits) gives: 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153.
#As the sum of the digits raised to the power of the number of digits equals the original number (153), it is an Armstrong
n=int(input())
num8, z = n, 0
while (n):  # counts the number of digits in n, which is 3
  n=n//10   #1) n=153//10= 15....z=1 2) n=15//10= 1...z=2  3) n=1//10= 0...z=3
  z+=1
sum1, k = 0, num8
while (num8):   # loop calculates the sum of each digit raised to the power of z, num8 is 153, and z is 3
  sum1+=(num8%10)**z # 1)(153%10)**3 calculates cube of last digit(3), which is added to sum1 2)(15%10)**3 calculates cube
                   #3) calculates the cube of the last digit (1), which is added to sum1
  num8=num8//10  # 1) num8 is updated to 15 after removing the last digit 2) num8 is updated to 1 3) num8 is updated to 0
if sum1==k:
  print("It is an armstrong number")
else:
  print("Not an armstrong number")
```

```
153
It is an armstrong number
```

```
num = 153
num1, sum = num, 0
while(num1!=0):
  sum += num1%10
  num1 //= 10
print(f"{sum}")
```

```
9
```

```
# Calculating Armstrong no for the no of times the user wants
t=int(input())
for i in range(t):
  num=int(input("Enter a Number: "))
  num1, sum = num, 0
  count = 0
  while(num1!=0):
    count+=1
    num1//=10
  num1 = num
  while(num1 != 0):
    sum += pow(num1%10, count)
    num1 //= 10
  if sum == num:
    print(f"{num} is an Armstrong Number")
  else:
    print(f"{num} is not an Armstrong Number")
```

```
2
Enter a Number: 123
123 is not an Armstrong Number
```

```
Enter a Number: 321
321 is not an Armstrong Number
```

```
#pass statement is a null operation. It doesn't do anything; it acts as a placeholder
#where Python syntax requires a statement, but no action needs to be taken.
#This is useful when you're in the process of designing your code structure
#or when you're writing a skeleton for future implementation.
for i in range(1, 5):
  print(i, "Before")
  #break
  #continue
  pass
  print(i, "After")
print(i, "After Loop")
```

```
1 Before
1 After
2 Before
2 After
3 Before
3 After
4 Before
4 After
4 After Loop
```

```
for i in range(1, 5):
  continue
  print(i, "IN")
print(i, "OUT")
```

```
4 OUT
```

```
n = 3
while n > 0:
  for x in range(3,10):
    if x%5 == n:
      print("Break Executed for Loop Terminated")
      break
    elif x%3 == n:
      print("Passing Here")
      pass
      print("Pass Executed Continue for loop without change")
    else:
      print("Continue Executed goto next iteration of for")
      continue
    print("Continue Not Executed")
  n -= 1
```

```
Break Executed for Loop Terminated
Continue Executed goto next iteration of for
Continue Executed goto next iteration of for
Passing Here
Pass Executed Continue for loop without change
Continue Not Executed
Continue Executed goto next iteration of for
Break Executed for Loop Terminated
Continue Executed goto next iteration of for
Passing Here
Pass Executed Continue for loop without change
Continue Not Executed
Continue Executed goto next iteration of for
Break Executed for Loop Terminated
```

```
#GCD of two numbers
x, y = input("Enter 2 Number: ").split()
x = int(x); y = int(y)

#x, y = [int(a) for a in input().split()]

#x, y = map(int, input().split())

if (y > x):
  x, y = y, x
while(y>0):
  x, y = y, x % y # 50, 40 --> 40, 10 --> 10, 0
print('GCD is: ', x)
```

```
Enter 2 Number: 56 32
GCD is:  8
```

```
e = [int(a) for a in input().split()]
e
```

```
12 55 45 78
[12, 55, 45, 78]
```

match Statement

“Switch Case in Python” called “Python Structure Pattern Matching”

match subject:

> **case <pattern_1>:**
>
> > **<action_1>**
>
> **case <pattern_2>:**
>
> > **<action_2>**
>
> **case <pattern_3>:**
>
> > **<action_3>**
>
> **case _:**
>
> > **<action_wildcard>**

```
!sudo apt-get install python3.10
!sudo update-alternatives --install /usr/bin/python3 python3 /usr/bin/python3.10 1
!sudo update-alternatives --set python3 /usr/bin/python3.10
```

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
  libnvidia-common-460
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libpython3.10-minimal libpython3.10-stdlib python3.10-minimal
Suggested packages:
  python3.10-venv binfmt-support
The following NEW packages will be installed:
  libpython3.10-minimal libpython3.10-stdlib python3.10 python3.10-minimal
0 upgraded, 4 newly installed, 0 to remove and 5 not upgraded.
Need to get 5,098 kB of archives.
After this operation, 19.5 MB of additional disk space will be used.
```